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Generative AI-assisted evaluation of ESG practices and information delays in ESG ratings

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ABSTRACT

We explore the time-lag effects of Environmental, Social, and Governance (ESG) practices' integration within firms from the NASDAQ 100 index, analyzing how this information is reflected over time through Generative AI. By employing ChatGPT-4 to assess ESG progress from 2011 to 2022 in their annual 10-K reports, we identify significant time-lagged effects between the AI-generated evaluations and the ESG ratings, particularly pronounced in the Environmental and Social pillars among firms with mid-range ESG ratings. Our evidence suggests that such information delay in ESG ratings is plausibly driven by reputational herding among firms in their ESG practices, along with potential greenwashing.

1. Introduction

With the growing importance of environmental concerns and the rise of sustainable development, public focus on corporate non-financial information has significantly intensified, driving investors to integrate ESG ratings into their investment processes (Van Duuren et al., 2016; Eccles et al., 2017; Vannoni and Ciotti, 2020). Beyond guiding investment decisions, ESG ratings play a crucial role in promoting green innovation and improving financial outcomes, such as reducing bond credit spreads by mitigating risks (Wang et al., 2023; Lian et al., 2023). However, ESG ratings may not be up-to-date or precise enough, as there can be delays in reflecting all the available information regarding the effective integration of the related practices. This phenomenon has already been observed in financial markets, with previous studies indicating that changes in credit ratings are often delayed and lag behind financial changes (Pinches and Singleton, 1978; Beaver et al., 2006; Cheng and Neamtiu, 2009; Even-Tov and Ozel, 2021). A similar latency could also exist in the integration of ESG practices.

Among firms with varying ESG development incentives, the rate of adoption of ESG practices vary significantly (Wang, 2023). Furthermore, the pace of ESG integration can differ not only among firms, but also within the specific categories of Environment, Social, and Governance. Such differences may arise because companies have varied incentives to enhance their ratings in each category (Pelosi and Adamson, 2016). As such, the challenges associated with interpreting ESG information can differ markedly across these areas (Paredes-Gazquez et al., 2014; Amel-Zadeh and Serafeim, 2018), leading to variations in the extent of information delays. Given the increasing importance of ESG ratings, understanding their effectiveness in accurately capturing and reflecting publicly available ESG disclosures is crucial.

To investigate this, we utilize Generative AI to analyze the potential time-lagged effects between publicly available ESG information and ESG ratings. The advancements in AI technology offer a promising avenue for a deeper exploration of textual data, thus

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Table 1
Companies from the NASDAQ 100 index.

ADOBE (NAS)	COMCAST A	MICRON TECHNOLOGY
ADVANCED MICRO DEVICES	COSTCO WHOLESALE	MICROSOFT
AMAZON.COM	CSX	MONDELEZ INTERNATIONAL CL.A
AMER.ELEC.PWR.	EXELON	NVIDIA
AMGEN	FASTENAL	ON SEMICONDUCTOR
ANALOG DEVICES	FORTINET	PACCAR
APPLE	HONEYWELL INTL.	PEPSICO
APPLIED MATS.	INTEL	QUALCOMM
ASML HLDG.ADR 1:1	INTUIT	ROPER TECHNOLOGIES
ASTRAZENECA ADR 2:1	KEURIG DR PEPPER	ROSS STORES
AUTODESK	KLA	STARBUCKS
BIOGEN	LAM RESEARCH	TESLA
BROADCOM	LULULEMON ATHLETICA	TEXAS INSTRUMENTS
CINTAS	MARRIOTT INTL.'A'	T-MOBILE US
CISCO SYSTEMS	MARVELL TECHNOLOGY	VERISK ANALYTICS CL.A
COGNIZANT TECH.SLTN.'A'	MICROCHIP TECH.	WARNER BROS DISCOVERY SERIES A
		XCEL ENERGY

enabling a more nuanced understanding of institutional evolution. Former studies such as those by Wyrobek (2020), Riskiyadi (2023), and Sultana et al. (2023) have successfully employed AI to address issues like unfair corporate culture or to predict future fraud. Additionally, AI has been applied in other domains, including forecasting firm performance, and enhancing financial reporting (Wamba-Taguimdje et al., 2020; Ledhem, 2021; Burçin Tutcu et al., 2024; Kim et al., 2024; Li et al., 2024). In this paper, we utilize ChatGPT-4 to independently assess the progression of ESG elements within annual 10-K reports. This choice is inspired by numerous studies demonstrating that the text analysis capabilities of large language models (LLMs), particularly ChatGPT-4, improve with increasing model size.[1] By integrating the AI-generated evaluations with ESG ratings from Refinitiv Datastream, we can also explore LLMs' capacity to discern and quantify genuine trends in ESG practices.

Two main conclusions are drawn from our results. First, the Environmental and Social pillars exhibit a notable time-lag effect, with significant correlations to AI-analyzed scores from the previous two years, indicating that the information related to ESG practices is gradually reflected in the corresponding ESG ratings over time. Second, firms with less developed ESG baselines integrate practices slower than those with advanced foundations, yet they exhibit shorter time lags compared to companies in the middle range.

In terms of possible interpretations, our findings indicate a tendency toward herding behavior in ESG development. Firms with similar initial ESG statuses often observe and replicate the ESG strategies of their peers. This emulation is especially prevalent among firms with less developed frameworks, who tend to follow the lead of more established peers. Research focusing on stock indices and banking has revealed herding behaviors in ESG development (Kerkemeier and Kruse-Becher, 2022; Danila, 2023; Wang, 2023), predominantly driven by reputational concerns (Scharfstein and Stein, 1990; Zwiebel, 1995). Also, information sourced from public domains tend to lack precision due to these herding dynamics, as firms attempt to mirror their peers without genuinely enhancing their ESG practices. As a result, ESG adoption activities motivated by herding may be less indicative of the actual outcomes expected from these implementations (Bikhchandani et al., 1992; Welch, 1992). The integration of these activities into ESG scores may be delayed due to the imprecision of the information resulting from information cascade. There has been extensive evidence that market prices and other quantitative measures, such as credit ratings, are slow to reflect changes when publicly available information is imprecise (Hou and Moskowitz, 2003; Löffler, 2005; Stice, 2023).

Greenwashing may also contribute to herding behaviors. Some studies have highlighted the discrepancy between ESG ratings and the actual policy advocacy reflected in filings, defining such discrepancies as greenwashing (Yu et al., 2020; Liao et al., 2023; Yuan et al., 2024). Firms with mid-range ESG ratings tend to be particularly prone to greenwashing due to their efforts to keep pace with their peers, reflecting herding behavior.

Additionally, AI faces certain limitations in evaluating companies' ESG disclosures. Unlike Refinitiv's quantitative, multi-metric ESG ratings, ChatGPT-4 depends on the quality of publicly available text data. This reliance may lead to biases due to selective or misleading disclosures (Mehrabi et al., 2019; Ferrara, 2023) and may overlook subtle signs of greenwashing. When companies exhibit herd behavior or engage in greenwashing tactics, the gap between manual and automated analyses in ESG integration may be exacerbated, causing ESG rating to lag actual disclosure progress. This lag is particularly pronounced in the Environmental and Social categories, where the pressure to meet high public expectations is greater compared to Governance standards. These insights align with the findings from our analysis.

The rest of this paper is organized as follows. Section 2 mainly introduces the research design of data and methods. Section 3 describes the empirical analysis process. Section 4 is the concluding remarks.

[1] Lopez-Lira and Tang (2023) found that basic models like GPT-1, GPT-2, and BERT have minimal predictive power for financial stock information. Additionally, more advanced models like GPT-3.5 and DistilBart-MNLI, although somewhat predictive, fall short of ChatGPT-4's ability to effectively interpret complex data. Even compared to the latest, commonly used models like Bard and Claude, ChatGPT-4 demonstrates superior text understanding and analytical accuracy (Borji & Mohammadian, 2023).

Table 2
Summary statistics of ESG ratings and AI-analyzed scores.

	Environmental Score		Social Score		Governance Score		AI-analyzed Environmental Image Score		AI-analyzed Social Image Score		AI-analyzed Governance Image Score	
Year	Mean	Std. err.	Mean	Std. err.	Mean	Std. err.	Mean	Std. err.	Mean	Std. err.	Mean	Std. err.
2011	54.498	26.695	58.130	20.930	58.616	25.449	63.469	14.113	66.837	12.653	76.551	6.038
2012	54.514	25.004	57.346	19.653	58.308	25.484	67.612	14.746	68.714	13.042	79.857	6.548
2013	54.652	24.622	60.422	18.457	61.064	25.204	68.204	13.569	70.939	11.586	79.918	6.487
2014	55.195	24.054	61.747	18.727	59.536	23.768	67.755	14.165	71.755	12.197	80.306	6.856
2015	57.047	23.112	63.676	16.995	63.698	19.537	68.449	14.283	71.857	12.086	79.735	7.053
2016	58.390	22.206	65.727	16.542	65.624	17.789	70.776	12.032	74.449	10.960	81.184	7.146
2017	60.832	21.595	66.246	17.983	65.136	18.175	71.959	11.025	73.449	11.264	81.367	7.607
2018	62.818	22.324	68.076	18.064	66.078	18.879	73.571	11.635	75.959	9.753	82.286	7.223
2019	67.847	18.474	69.848	17.522	64.965	18.503	75.571	9.910	76.571	10.271	83.020	7.341
2020	68.969	17.061	72.433	16.509	66.182	18.870	77.469	9.685	80.204	7.695	84.449	5.831
2021	70.722	14.819	73.271	15.951	67.385	15.366	80.449	8.851	84.143	5.759	86.449	4.809
2022	71.551	12.911	74.153	13.715	69.700	15.214	82.980	6.796	85.878	4.644	87.224	4.515
All years	61.419	22.227	65.923	18.375	63.858	20.618	72.355	13.121	75.063	11.814	81.862	7.088

The table presents descriptive statistics of Environmental, Social, and Governance scores alongside AI-analyzed scores. Mean and standard error values are reported for each year between 2011 and 2022.

2. Data and methodology

2.1. Data

We consider a sample of N = 49 companies from the NASDAQ 100 index,[2] excluding companies with incomplete data within the observed period. The period covers from 2011 to 2022 (T= 12), with data sourced from Refinitiv Datastream (Table 1). The collected dataset comprises yearly Environmental, Social, and Governance scores. Next, we manually downloaded their annual reports and analyzed the performance of each ESG pillar (E, S, and G separately) using ChatGPT-4, which assigned a score on a 100-point scale. To ensure scoring consistency, we used each company's annual 10-K reports[3] as the sole data source. For the scoring process, each company was evaluated within an independent window, using its annual 10-K reports from 2011 to 2022, in sequential order. Following the method used in previous research (Romanko et al., 2023; Ko and Lee, 2024), we provided a specified prompt to support the evaluation.[4]

2.2. Empirical methodology

To examine the relationship between historical AI-analyzed scores and current ESG ratings, separate regression models are used for each ESG pillar. The baseline model is formulated as follows to evaluate how AI-analyzed scores from up to five previous years influence the current year's ratings:

$$\text{PillarScore}_{i,t} = \alpha + \beta_1 \text{AIScore}_{i,t-1} + \beta_2 \text{AIScore}_{i,t-2} + \beta_3 \text{AIScore}_{i,t-3} + \beta_4 \text{AIScore}_{i,t-4} + \beta_5 \text{AIScore}_{i,t-5} + \mu_i + \epsilon_{it}$$

In this model, $\text{PillarScore}_{i,t}$ represents the current score for the Environmental, Social, or Governance pillar for company i in year t. $\text{AIScore}_{i,t-k}$ denotes the AI-analyzed score for the respective pillar for company i at time $t-k$ for k from 1 to 5, capturing the ESG performance as reflected in their disclosures from one to five years prior. α represents the intercept of the model, β_k are the coefficients for the lagged AI-analyzed pillar scores, μ_i represents the random effects that capture time-invariant company-specific characteristics, while ε_{it} is the idiosyncratic error term.

Based on the theoretical background discussed in the introduction, we propose the following hypotheses:

Hypothesis 1. Time-lagged effects exist in the adoption of Environmental, Social, and Governance practices.

Hypothesis 2. The speed of ESG integration varies among companies based on their different foundational stages of ESG development.

[2] The NASDAQ 100 index includes some of the most innovative and influential companies in the technology and biotech sectors, which are increasingly central to global economic trends and sustainability issues, making their ESG practices particularly valuable for analysis.

[3] Mandated by the U.S. Securities and Exchange Commission (SEC), 10-K reports follow rigorous disclosure standards and a standardized format, providing reliable and comparable information across all companies and years.

[4] "Can you help me assess its ESG development for that year based on the annual report and give a score on a 100-point scale (separately for Environmental, Social, Governance)?"

Table 3
Regression analysis of lagged AI-analyzed ESG scores and current ESG performance.

Dep. Var.: Refinitiv Environmental Score						
Environmental Pillar	AI-analyzed Environmental Image Score (−1)	0.628***	0.527***	0.466***	0.515***	0.383***
		(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
	AI-analyzed Environmental Image Score (−2)		0.272***	0.228***	0.245***	0.312***
			(0.000)	(0.004)	(0.002)	(0.000)
	AI-analyzed Environmental Image Score (−3)			0.207***	0.171**	0.223***
				(0.006)	(0.029)	(0.006)
	AI-analyzed Environmental Image Score (−4)				0.020	0.001
					(0.784)	(0.994)
	AI-analyzed Environmental Image Score (−5)					0.045
						(0.546)
Dep. Var.: Refinitiv Social Score						
Social Pillar	AI-analyzed Social Image Score (−1)	0.475***	0.327***	0.256***	0.229***	0.179***
		(0.000)	(0.000)	(0.000)	(0.000)	(0.005)
	AI-analyzed Social Image Score (−2)		0.263***	0.208***	0.165**	0.190***
			(0.000)	(0.002)	(0.016)	(0.005)
	AI-analyzed Social Image Score (−3)			0.189***	0.152**	0.146**
				(0.005)	(0.027)	(0.033)
	AI-analyzed Social Image Score (−4)				0.130**	0.119*
					(0.044)	(0.067)
	AI-analyzed Social Image Score (−5)					−0.010
						(0.869)
Dep. Var.: Refinitiv Governance Score						
Governance Pillar	AI-analyzed Governance Image Score (−1)	0.539***	0.456***	0.365***	0.293**	0.298**
		(0.000)	(0.000)	(0.004)	(0.015)	(0.014)
	AI-analyzed Governance Image Score (−2)		0.201	0.199	0.110	0.122
			(0.121)	(0.141)	(0.401)	(0.350)
	AI-analyzed Governance Image Score (−3)			0.235*	0.227*	0.204
				(0.072)	(0.090)	(0.130)
	AI-analyzed Governance Image Score (−4)				0.174	0.204
					(0.187)	(0.153)
	AI-analyzed Governance Image Score (−5)					−0.077
						(0.583)

The table reports regression results evaluating the relationship between lagged AI-analyzed ESG scores and current Refinitiv ESG ratings. Columns detail coefficients and significance levels for different lag intervals (1–5 years) across Environmental, Social, and Governance pillars. *, **, *** indicate confidence levels of 10 %, 5 %, and 1 %, with p-values in parentheses.

3. Empirical results

Table 2 presents the summary statistics for the key variables used in our regression analysis. The Environmental scores have significantly increased, rising from an average of 54.498 in 2011 to 71.551 in 2022. Similarly, Social scores have increased from 58.13 to 74.153, accompanied by a decrease in standard error, indicating a growing emphasis on social engagement and a convergence of Environmental and Social performance indicators across companies. The AI-analyzed image scores for all pillars also show a consistent upward trend, with particularly notable increases in the Environmental pillar from 63.469 to 82.98 and in the Social pillar from 66.837 to 85.878. Subsequently, we also conducted a correlation analysis of the main variables, as shown in Table A.1 (Appendix). Overall, the current Refinitiv scores exhibit a certain correlation with AI-analyzed scores within the 1-year to 3-year lags. However, starting from a 4-year lagged variable, the correlation coefficients display a marked decrease. For example, regarding the Environmental pillar, the correlation coefficient decreases from 0.503 at a 1-year lag to approximately 0.42 at a 4-year lag. Similarly, for the Social pillar, the correlation coefficient drops by about 7 percentage points from a 1-year lag to a 4-year lag.

The regression results reveal a significant positive correlation across all pillars (Table 3), particularly pronounced with the 1-year and 2-year lagged variables, underscoring a notable time-lag effect in ESG practices. In the Environmental pillar, AI-analyzed scores from one and two years prior (t-1, t-2) show strong correlations with current Refinitiv ratings, with coefficients of 0.628 and 0.527 for the first two regressions, both significant at the 1 % level. Similarly, the Social pillar shows a robust initial correlation (t-1) at 0.475 ($p<$ 0.01), which declines over time but remains significant up to a 3-year lag. By contrast, the Governance pillar exhibits a quicker integration of practices, with significant initial lag correlations that diminish more rapidly than in the Environmental and Social pillars.

Environmental and Social initiatives typically require complex, resource-intensive changes that are difficult to implement and seldom yield measurable results quickly. For instance, modifying corporate environmental practices might involve extensive infrastructural changes or innovative technology implementations, both of which are costly and time-consuming (Albort-Morant et al., 2016; Fosu et al., 2023). Social initiatives often require long-term investments in community engagement or shifts in organizational culture (Pelosi and Adamson, 2016). However, Governance reforms involving policy or compliance updates can be implemented more swiftly as they generally do not require the broad logistical changes needed for Environmental or Social reforms.

To further examine the temporal dynamics of ESG practices, we categorized companies into quartiles based on their initial 2011

Table 4(a)
Quartile-based analysis on environmental pillar.

Dep. Var.: Refinitiv Environmental Score				
	AI-analyzed Environmental Image Score (−1)	−0.040	0.094	0.151
		(0.643)	(0.425)	(0.236)
Over 75 % Percentiles	AI-analyzed Environmental Image Score (−2)		−0.126	−0.118
			(0.211)	(0.379)
	AI-analyzed Environmental Image Score (−3)			−0.088
				(0.439)
	AI-analyzed Environmental Image Score (−1)	0.360***	0.240**	0.212*
		(0.000)	(0.013)	(0.063)
50–75 % Percentiles	AI-analyzed Environmental Image Score (−2)		0.180*	0.109
			(0.057)	(0.360)
	AI-analyzed Environmental Image Score (−3)			0.129
				(0.195)
	AI-analyzed Environmental Image Score (−1)	0.358***	0.347***	0.341***
		(0.000)	(0.001)	(0.002)
25–50 % Percentiles	AI-analyzed Environmental Image Score (−2)		0.193*	0.219**
			(0.058)	(0.044)
	AI-analyzed Environmental Image Score (−3)			0.101
				(0.365)
	AI-analyzed Environmental Image Score (−1)	0.989***	0.793***	0.721***
		(0.000)	(0.000)	(0.001)
Under 25 % Percentiles	AI-analyzed Environmental Image Score (−2)		0.342*	0.164
			(0.097)	(0.482)
	AI-analyzed Environmental Image Score (−3)			0.366*
				(0.077)

Table 4(b)
Quartile-based analysis on social pillar.

Dep. Var.: Refinitiv Social Score				
	AI-analyzed Social Image Score (−1)	−0.076	−0.133	−0.163
		(0.414)	(0.226)	(0.133)
Over 75 % Percentiles	AI-analyzed Social Image Score (−2)		0.143	0.156
			(0.196)	(0.169)
	AI-analyzed Social Image Score (−3)			0.188*
				(0.094)
	AI-analyzed Social Image Score (−1)	0.572***	0.236	0.172
		(0.000)	(0.110)	(0.253)
50–75 % Percentiles	AI-analyzed Social Image Score (−2)		0.500***	0.249
			(0.002)	(0.141)
	AI-analyzed Social Image Score (−3)			0.361**
				(0.027)
	AI-analyzed Social Image Score (−1)	0.466***	0.315***	0.155*
		(0.000)	(0.000)	(0.081)
25–50 % Percentiles	AI-analyzed Social Image Score (−2)		0.263***	0.231**
			(0.003)	(0.016)
	AI-analyzed Social Image Score (−3)			0.258***
				(0.005)
	AI-analyzed Social Image Score (−1)	0.569***	0.506***	0.524***
		(0.000)	(0.000)	(0.000)
Under 25 % Percentiles	AI-analyzed Social Image Score (−2)		0.175	0.187
			(0.221)	(0.208)
	AI-analyzed Social Image Score (−3)			0.011
				(0.943)

Refinitiv ESG scores: under 25 %, 25–50 %, 50–75 %, and over 75 %. Analysis shows that companies in the lower percentiles (under 25 % and 25–50 %) maintain significant positive correlations with Refinitiv scores across multiple lags, especially in the Environmental and Social pillars. Specifically, the 25–50 % group demonstrates significant correlations in Social scores for up to three years, suggesting a slow integration of social practices. Conversely, companies in the highest percentile (over 75 %) exhibit weaker significance levels at longer lags, indicating that more developed firms can swiftly implement new policies with only minor adjustments, thus experiencing shorter time lags (Tables 4a-4c).

Based on these results, to gain deeper insight into how ChatGPT-4′s ESG evaluations align or diverge from Refinitiv's ratings over time, we created a time series plot to validate these trends (Fig. 1). First, we structured the data by establishing a six-year rolling time window, in which each window includes six years of ESG scores for all companies. For each overlapping relative year, companies were divided into three groups based on their AI-analyzed scores: (1) the top 20 %, (2) the middle 40 % to 60 %, and (3) the bottom 20 %. We

Table 4(c)
Quartile-based analysis on governance pillar.

Dep. Var.: Refinitiv Governance Score				
	AI-analyzed Governance Image Score (−1)	−0.074	0.193	0.343
		(0.703)	(0.415)	(0.125)
Over 75 % Percentiles	AI-analyzed Governance Image Score (−2)		−0.031	0.152
			(0.880)	(0.499)
	AI-analyzed Governance Image Score (−3)			−0.306
				(0.123)
	AI-analyzed Governance Image Score (−1)	−0.160	0.032	−0.048
		(0.539)	(0.922)	(0.889)
50–75 % Percentiles	AI-analyzed Governance Image Score (−2)		−0.244	−0.068
			(0.446)	(0.859)
	AI-analyzed Governance Image Score (−3)			0.300
				(0.393)
	AI-analyzed Governance Image Score (−1)	0.683***	0.551***	0.557***
		(0.000)	(0.006)	(0.007)
25–50 % Percentiles	AI-analyzed Governance Image Score (−2)		0.281	0.057
			(0.187)	(0.796)
	AI-analyzed Governance Image Score (−3)			0.398*
				(0.072)
	AI-analyzed Governance Image Score (−1)	0.924***	0.533*	0.230
		(0.000)	(0.055)	(0.371)
Under 25 % Percentiles	AI-analyzed Governance Image Score (−2)		0.511*	0.396
			(0.089)	(0.175)
	AI-analyzed Governance Image Score (−3)			0.273
				(0.337)

The tables present the regression results for Environmental, Social, and Governance pillars, segmented by quartiles of companies' initial ESG scores (under 25 %, 25–50 %, 50–75 %, and over 75 %). Each table evaluates the relationship between lagged AI-analyzed scores and current Refinitiv ratings within quartile groups. *, **, *** indicate confidence levels of 10 %, 5 %, and 1 %, with p-values in parentheses.

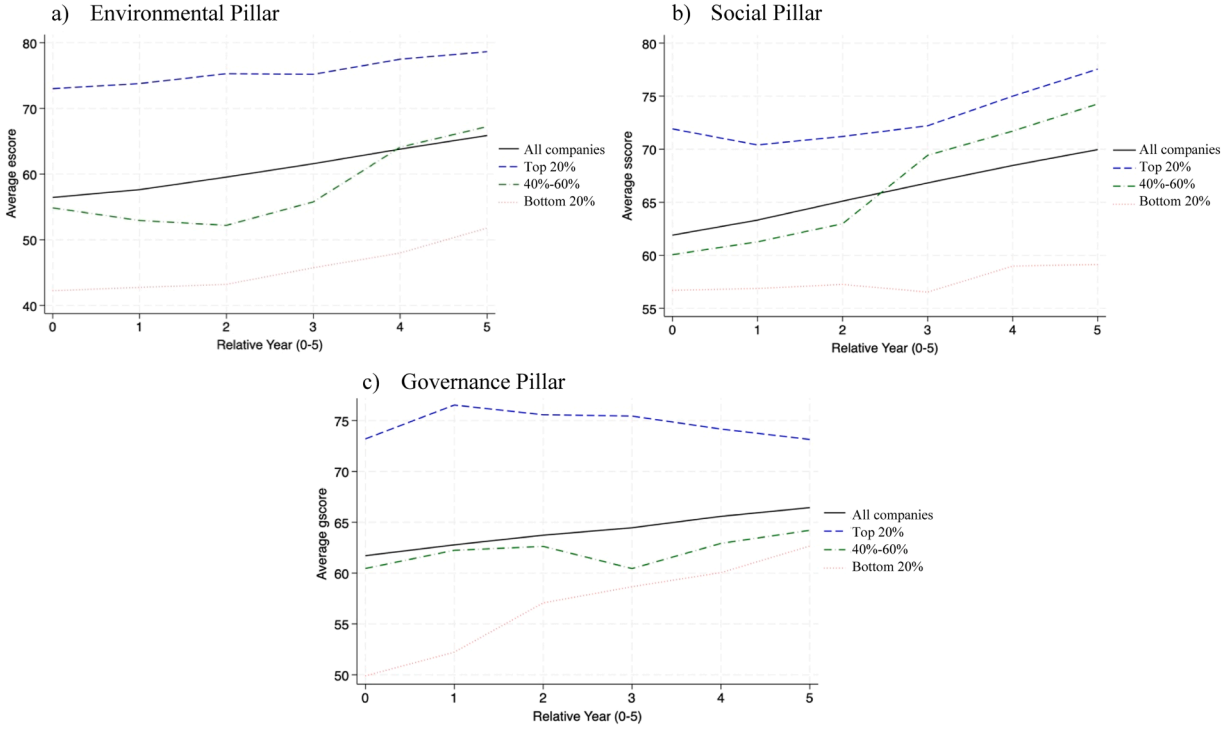


Fig. 1. Average refinitiv ESG scores by AI-analyzed score percentiles.
The figure illustrates the temporal dynamics of Refinitiv ESG scores across percentile-based groups of AI-analyzed scores within a six-year rolling time window. Companies are categorized into three groups based on their AI-analyzed ESG scores: the top 20 %, middle 40–60 %, and bottom 20 %. For each group, the mean Refinitiv ESG score is calculated and plotted over a relative time span, ranging from Year 0 (the current evaluation year) to Year 5 (up to five subsequent years).

then calculated the average Refinitiv score for each group, resulting in a mean series across years 0 to 5. The chart reveals that for the Environmental and Social pillars, the Refinitiv rating exhibits a more pronounced increase after a lag of two to three years, especially notable in the trend line of mid-range companies. This observation implies that ESG disclosure information may require approximately two to three years to be fully integrated and reflected in actual ESG practices.

We suppose that the observed delays in ESG score integration could also be attributed to herding behavior, where firms replicate the ESG practices of their peers. This imitation could result in lower-quality ESG information because the tendency of herding prevents further revelation of useful information. This phenomenon corresponds to the concept of informational cascades, where firms overlook potentially valuable private information in favor of copying others, particularly in uncertain environments (Bikhchandani et al., 1992; Welch, 1992). Moreover, as companies herd in their ESG implementations, available information on their activities becomes more limited, resulting in decreased effectiveness of information gathering (Han and Sangiorgi, 2018). This imprecision can not only impair rating agencies' ability to accurately reflect true performance, but also reduce their incentives to gather more information.[5] This argument is consistent with broader financial literature, which shows that market responses are delayed when information is vague or unreliable. Similar trends are observed in slow price adjustments (Hou and Moskowitz, 2003; Stice, 2023) and credit rating changes (Löffler, 2005).

AI may also face certain challenges in evaluating the ESG disclosures. Unlike Refinitiv's quantitative, multi-metric ESG ratings, ChatGPT-4 heavily relies on the quality of publicly available text data, which may perpetuate biases reflected in selective or misleading disclosure data (Mehrabi et al., 2019; Ferrara, 2023), and may overlook subtle indicators of greenwashing behavior. Therefore, for companies prone to greenwashing, the gap in ESG integration feedback between manual and automated analyses may widen further, as these companies might exaggerate their ESG disclosures to appear proactive without genuinely implementing substantial changes (Yu et al., 2020; Liao et al., 2023; Yuan et al., 2024). Such misleading ESG disclosures are often intended to keep up with industry peers through herding behavior, ultimately causing ESG ratings to lag behind the actual disclosures. This is also evidenced in the finding, firms in the mid-range quartile demonstrate prolonged correlations in Environmental and Social scores, indicating a heightened susceptibility to greenwashing. In other words, the extended adjustment period reflects a disconnect between signaling and actual practices.

4. Concluding remarks

Our research is a novel attempt to assess corporate ESG evolution through advanced AI tools. We investigate the time-lag effects and precision of ESG ratings using AI-based analysis. By applying ChatGPT-4 to evaluate corporate ESG disclosures, we find that the integration of ESG practices into ratings occurs gradually, particularly in the Environmental and Social categories. This delay is particularly evident among firms with less mature ESG frameworks, who tend to follow the practices of their peers, reflecting a tendency of herding.

These findings highlight the potential utility of ChatGPT-4 in advancing ESG reporting. When accuracy is ensured, ChatGPT-4 can expedite the extraction of ESG information from reports, allowing agencies to benefit from more up-to-date information. However, companies with mid-range ESG ratings may be more susceptible to greenwashing behaviors, potentially due to herding effects, which can result in significant noisy information. Therefore, it would require more effort to disentangle noise from true information. Effectively identifying and integrating data from diverse sources warrants further research and requires special attention from rating agencies and investors.

CRediT authorship contribution statement

Qishu Wang: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation.

Conflicts of Interest Statement

The authors whose names are listed immediately below certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

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[5] There is evidence that more precise public information such as higher media coverage for firms can enhance their ESG performance (He et al., 2023). This indeed supports the idea that a lack of precise information can result in a misalignment of incentives in ESG performance.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.frl.2025.106757.

Data availability

Data will be made available on request.

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